

SEAT No. _____

[20]

SARDAR PATEL UNIVERSITY

M.Sc. IT (Semester-1) Examinations - 2021
 PS01CINT33 : Computer Networks & Cyber Security
 Wednesday, 24th February, 2021

Total Marks: 70

Time: 10.00 am to 12.00 pm

[08]

Q.1[A]

Give answers of following Multiple Choice Questions:

- i. OSI stands for _____.
 (A) Open Systems Interconnection (B) Open Service Interconnection
 (C) Open Systems Internet (D) None of these
- ii. Which of the following is not a guided media?
 (A) Twisted-pair cable (B) Fiber-optic cable
 (C) Coaxial cable (D) Atmosphere
- iii. The signal from a satellite is normally aimed at a specific area called _____.
 (A) Channel (B) Footprint
 (C) Tower (D) Orbit
- iv. In _____ routing technique, every incoming packet is sent out on every outgoing line except the one it arrived on.
 (A) Dynamic (B) Distant Vector
 (C) Flooding (D) Shortest Path
- v. In all mobile phone systems, a geographic region is divided up into _____.
 (A) Packets (B) Bits
 (C) Cells (D) Streams
- vi. A network built up from company computer and leased telephone lines is called _____.
 (A) Public network (B) Private network
 (C) General network (D) None
- vii. The general system of symbol-for-symbol substitution is called a _____.
 (A) Monoalphabetic (B) Alphabetic
 (C) Arithmetic (D) None
- viii. _____ is a process of recovering passwords from data that have been stored in or transmitted by a computer system.
 (A) Password cracking (B) Forgery
 (C) Phishing (D) Spamming

[16]

Q-1[B]

Do as directed:

- i. Which layer of TCP/IP model has the functionality same as the Network layer of OSI model?
- ii. The _____ communication mode support two-way traffic but in only one direction at a time.
- iii. When too many packets are present in (a part of) the subnet, performance degrades. This situation is called _____.
- iv. What is routing algorithm?
- v. What do you mean by frequency reusability?
- vi. Define Fragmentation.
- vii. The output of the encryption process, known as the _____.
- viii. Full form of DDOS is _____.
- ix. Framing is the responsibility of Data link layer. [TRUE/FALSE]

- x. A repeater operates at Network layer. [TRUE/FALSE]
- xi. Timeout determination policy is methods to control congestion at Transport layer. [TRUE/FALSE]
- xii. Gigabit Ethernet supports 100 Mbps speed. [TRUE/FALSE]
- xiii. DES stands for Data Encryption situation. [TRUE/FALSE]
- xiv. AMPS stands for _____.
- xv. What is key loggers?
- xvi. Cyber security refers to sending a large number of E-Mails to the victim to crash victim's E-Mail account (in the case of an individual) or to make victim's mail servers crash. [TRUE/FALSE]

Q.2 Answer the following Short Questions: (Attempt any Seven) [14]

- i. Explain the function of Router.
- ii. List Unguided transmission media. What are the properties of Radio waves?
- iii. Define satellite orbit.
- iv. Differentiate between Non adaptive(Static routing) algorithms and Adaptive (Dynamic routing) algorithms.
- v. What is Tunneling?
- vi. Differentiate between soft handoff and hard handoff.
- vii. List Various classifications of cybercrimes.
- viii. What is cryptology?
- ix. What are the advantages of Fiber Optics?

- Q.3 [A] Write differences between LAN, MAN and WAN. [4]**
[B] Explain Twisted Pair cable in detail. [4]

OR

- Q-3 Discuss OSI Reference model. [8]**

- Q.4 [A] Write a note on geosynchronous satellite. [4]**
[B] What is Congestion? Write the reasons for getting congestion. [4]

OR

- Q.4 List Routing techniques. Explain Shortest Path Routing with example. [8]**

- Q.5 [A] Explain transparent fragmentation. [4]**
[B] Write a short note on Virtual Private Network with diagram. [4]

OR

- Q-5 [A] Explain GSM—The Global System for Mobile Communications. [4]**
[B] Discuss the factors on which networks differ. [4]

- Q.6 [A] Explain Substitution Ciphers with example. [4]**
[B] Write a short note on Spamming. [4]

OR

- Q.6 [A] What is firewall? Explain Packet filter firewall in detail With diagram. [4]**
[B] Explain a) Hacking and b) Credit Card Frauds [4]